

Chenxu Zhao

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Education

Iowa State University

Ames, IA

Ph.D. in Computer Science

Jan. 2023 – May. 2027 (Expected)

The Chinese University of Hong Kong, Shenzhen

Shenzhen, China

B.S. in Statistics

Aug. 2018 – May 2022

Internship Experience

Amazon

Bellevue, WA

Applied Scientist Intern, Agent Evaluation & Self-Evolving Agents

Jun. 2026 – Sep. 2026

- **Phase 1: Evaluation hardening.** Audited and hardened a production, multi-criterion LLM-as-a-judge system for a deployed GenAI agent across stability, discriminativeness, monotonicity, actionability, groundedness, and calibration; implemented **3** evaluator updates over **4** testing rounds, reducing verdict instability on a key criterion by **9×**.
- **Phase 2: Agent self-improvement.** Built an autonomous self-improvement loop for the production GenAI agent using the hardened evaluator as a multidimensional fitness signal; integrated three dedicated multi-agent workflow harnesses (*Explore*, *Diagnose*, *Propose*) with a variance harness, tree-structured evolutionary search, Pareto optimization, and deterministic validation/retention gates.
- **Scope and results.** Enabled controlled self-improvement *across prompts, tool descriptions, knowledge-base content, and bounded code regions*; ran the end-to-end loop unattended for **16** hours, orchestrating **579** subagent executions. Achieved **+14.4** pp, **+7.9** pp, and **+18** pp across three key criteria with no regressions beyond measured noise.

Selected Research Projects

Large Language Model Unlearning Security Vulnerabilities

- Built **LUSB**, a security benchmark for LLM unlearning that formalizes attacks across four threat scenarios and evaluates **16** attack/defense methods, **13** LLM architectures (up to 70B), **12** datasets, and **9** unlearning algorithms.
- Designed subset-optimization attacks that achieved up to **3×** improvement over baseline attacks.

Vision-Language Model Uncertainty Quantification

- Developed a reasoning-answer uncertainty framework with **finite-sample guarantees** and a **Shapley-value-based** module for attributing uncertainty to data and reasoning steps.
- Validated on **3 VLM architectures** across challenging multimodal benchmarks; published at **ACL 2026**.

Large Language Model Unlearning Privacy Vulnerabilities

- Constructed **PrivUB**, benchmarking **21** attacks/defenses across **11** datasets, **10** models, and **10** unlearning algorithms for membership inference, reconstruction, and knowledge leakage.
- Showed that perturbing unlearned data increased recovered-knowledge ROUGE by **79%** on Llama-3-8B/RWKU with RMU; published at **AAAI 2026 (Oral)**.

Membership Inference Attacks with Reliable Control

- Proposed **MIAFdR**, a conformal-inference wrapper that provides **provable false discovery rate control** for existing membership inference attacks.
- Validated across **3** datasets, **4** architectures, and **5+** MIA families with $< 0.01\%$ runtime overhead; published at **ICCV 2025**.

Selected Publications [Google Scholar]

Towards Benchmarking Privacy Vulnerabilities in Selective Forgetting with LLMs. Wei Qian, **Chenxu Zhao**, Yangyi Li, Mengdi Huai. *AAAI 2026, Oral*.

Membership Inference Attacks with False Discovery Rate Control. **Chenxu Zhao**, Wei Qian, Aobo Chen, Mengdi Huai. *ICCV 2025*.

Rethinking Adversarial Robustness in the Context of the Right to be Forgotten. **Chenxu Zhao**, Wei Qian, Yangyi Li, Aobo Chen, Mengdi Huai. *ICML 2024*.

Static and Sequential Malicious Attacks in the Context of Selective Forgetting. **Chenxu Zhao**, Wei Qian, Rex Ying, Mengdi Huai. *NeurIPS 2023*.